Dai Yuxuan

3036292116,

Comp Sc

MATH 3603–Homework 1

Due: 23-Sep-2026

1. Prove the following formula using induction:

$$P(E_1 \cup E_2 \cup \cdots \cup E_n) = \sum_i P(E_i) - \sum_{i<j} P(E_i \cap E_j) + \sum_{i<j<k} P(E_i \cap E_j \cap E_k)$$

$$- \sum_{i<j<k<l} P(E_i \cap E_j \cap E_k \cap E_l) + \cdots + (-1)^{n+1} P(E_1 \cap E_2 \cap \cdots \cap E_n).$$

2. For events E_1, E_2, $\cdots$, E_n, show that

$$P(E_1 \cap E_2 \cap \cdots \cap E_n) = P(E_1)P(E_2|E_1)P(E_3|E_1 \cap E_2) \cdots P(E_n|E_1 \cap \cdots \cap E_{n-1}).$$

3. Prove Boole's inequality:

$$P(\cup_{i=1}^n E_i) \leq \sum_{i=1}^n P(E_i).$$

hint: First consider the simple case when $n = 2$. The idea is to try to partition $\cup_{i=1}^n E_i$ into disjoint (mutually exclusive) set F_i's, where $F_1 = E_1$ and $F_i = E_i \cap (\cap_{j=1}^{i-1} (E_j)^c)$.

4. Only two factories produce rulers. 20 percent of the rulers from factory I and 5 percent from factory II are defective. Factory I produces twice as many rulers as factory II each week.

(i): What is the probability that a ruler, randomly chosen from a week's production, is satisfactory?

(ii): If the chosen ruler is defective, what is the probability that it came from factory I?

5. If events A, B, C are independent, prove the following statements

(i) A is independent of B^c,

(ii) A is independent of $B \cap C$,

(iii) A is independent of $B \cup C$.

P1.

Base case: We verify that when $n=2$, it holds

i.e. $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$.

proof: Note that: $E_1 \cup E_2 = (E_1 \setminus E_2) \cup (E_1 \cap E_2) \cup (E_2 \setminus E_1)$

and $E_1 = (E_1 \setminus E_2) \cup (E_1 \cap E_2)$, $E_2 = (E_1 \cap E_2) \cup (E_2 \setminus E_1)$

also $(E_1 \setminus E_2)$, $(E_1 \cap E_2)$, $(E_2 \setminus E_1)$ are pirewise mutually exclusive.

$$\therefore P(E_1 \cup E_2) = P(E_1 \setminus E_2) + P(E_1 \cap E_2) + P(E_2 \setminus E_1)$$
$$= P(E_1 \setminus E_2) + P(E_1 \cap E_2) + P(E_1 \cap E_2) + P(E_2 \setminus E_1) - P(E_1 \cap E_2)$$
$$= P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

Q.E.D.

Inductive step: Assume that $n=k \geq 2$ holds, then we show $n=k+1$ also holds.

proof: $P(\bigcup_{i=1}^{k+1} E_i) - RHS$ (where RHS is the form we want to show.)

$$= P\left(\left(\bigcup_{i=1}^{k} E_i\right) \cup E_{k+1}\right) - RHS$$
$$= P\left(\bigcup_{i=1}^{k} E_i\right) + P(E_{k+1}) - P\left(\left(\bigcup_{i=1}^{k} E_i\right) \cap E_{k+1}\right) - RHS$$
$$= P\left(\bigcup_{i=1}^{k} E_i\right) + P(E_{k+1}) - RHS - P\left(\bigcup_{i=1}^{k} (E_i \cap E_{k+1})\right)$$
$$= -\sum_{i=1}^{k} P(E_i \cap E_{k+1}) + \sum_{i<j\leq k} P(E_i \cap E_j \cap E_{k+1}) - \cdots + (-1)^k P\left(\bigcap_{i=1}^{k+1} E_i\right) - P\left(\bigcup_{i=1}^{k} (E_i \cap E_{k+1})\right)$$

$= 0$ (By expand the $P\left(\bigcup_{i=1}^{k} (E_i \cap E_{k+1})\right)$)

i.e. $P(\bigcup_{i=1}^{k+1} E_i) = RHS$. Q.E.D

Therefore, $n=k+1$ holds if $n=k$ holds.

Conclusion: By the PMI, it holds for $\forall n \geq 2$.

P2. Note that $P(E_k \mid \bigcap_{i=1}^{k-1} E_i) = \dfrac{P(\bigcap_{i=1}^{k+1} E_i)}{P(\bigcap_{i=1}^{k} E_i)}$

$$\therefore \text{RHS} = P(E_1) \cdot \prod_{k=2}^{n} \frac{P(\bigcap_{i=1}^{k} E_i)}{P(\bigcap_{i=1}^{k-1} E_i)} = \frac{\prod_{k=1}^{n} P(\bigcap_{i=1}^{k} E_i)}{\prod_{k=1}^{n-1} P(\bigcap_{i=1}^{k} E_i)} = P(\bigcap_{i=1}^{n} E_i) = \text{LHS}.$$

Q.E.D.

P3. $\forall n \geq 2 \in \mathbb{N}^+$.

$$\begin{aligned} P(\bigcup_{i=1}^{n} E_i) &= P(\bigcup_{i=1}^{n-1} E_i \cup E_n) \\ &= P(\bigcup_{i=1}^{n-1} E_i) + P(E_n) - P(\bigcup_{i=1}^{n-1} E_i \cap E_n) \\ &\leq P(\bigcup_{i=1}^{n-1} E_i) + P(E_n) \\ &\leq P(\bigcup_{i=1}^{n-2} E_i) + P(E_{n-1}) + P(E_n) \\ &\cdots \\ &\leq \sum_{i=1}^{n} P(E_i) \end{aligned}$$

4. Only two factories produce rulers. 20 percent of the rulers from factory I and 5 percent from factory II are defective. Factory I produces twice as many rulers as factory II each week.

(i): What is the probability that a ruler, randomly chosen from a week's production, is satisfactory?

(ii): If the chosen ruler is defective, what is the probability that it came from factory I?

P4.

We set those events: Randomly choose a ruler from a week's production:

A: From factory I; B: From factory II.

S: The ruler is satisfactory D: The ruler is defective.

Obviously (A, B) and (S, D) are all mutually exclusion and $P(A+B) = P(S+D) = 1$. $P(S|A) = 0.8$, $P(S|B) = 0.95$, $P(A) = \frac{2}{3} = 2P(B)$

(i) $P(S) = P(S \cap A) + P(S \cap B) = P(S|A)P(A) + P(S|B)P(B)$

$= 0.8 \times \frac{2}{3} + 0.95 \times \frac{1}{3} = 0.85$

(ii) By Bayes's rule: $P(A|D) = \dfrac{P(A)P(D|A)}{P(A)P(D|A) + P(B)P(D|B)} = \dfrac{\frac{2}{3} \times 0.2}{\frac{2}{3} \times 0.2 + \frac{1}{3} \times 0.05} = \frac{8}{9}$

P6.

(i) $P(A) = P(A \cap S)$

$= P(A \cap (B \cup B^c))$

$= P((A \cap B) \cup (A \cap B^c))$

$= P(A \cap B) + P(A \cap B^c) - 0$

$\therefore P(A \cap B^c) = P(A) - P(A \cap B)$

$= P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B^c)$

Q.E.D.

(ii) $P(A \cap (B \cap C)) = P(A \cap B \cap C) = P(A)P(B)P(C)$ (Definition of mutual independence)

$= P(A) \cdot P(B \cap C)$

(iii) $P(A \cap (B \cup C)) = P((A \cap B) \cup (A \cap C))$

$= P(A \cap B) + P(A \cap C) - P((A \cap B) \cap (A \cap C))$

$= P(A)P(B) + P(A)P(C) - P(A)P(B)P(C)$

$= P(A)(P(B) + P(C) - P(B)P(C))$

$= P(A)(P(B) + P(C) - P(B \cap C))$

$= P(A)P(B \cup C)$